

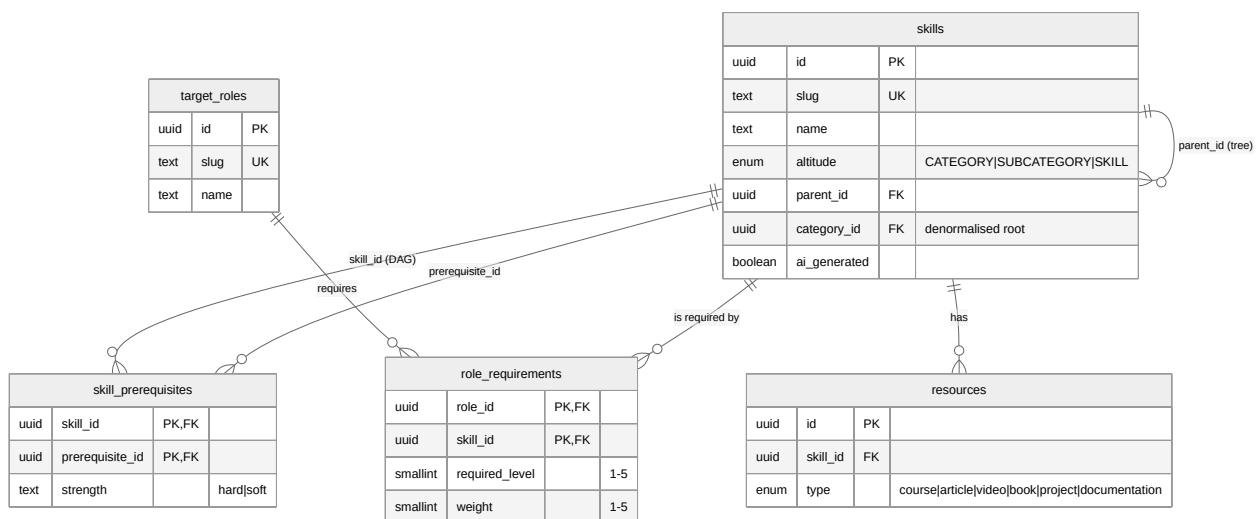
Database schema

PostgreSQL 17 with pgvector. One database, one migration history in `packages/db/migrations`, applied by the `migrate` container before any service starts.

The shape in one paragraph

`skills` is one table with an `altitude` enum (CATEGORY / SUBCATEGORY / SKILL). Two structures sit over it: a **tree** via `parent_id` (the Skill Tree view) and a **DAG** via `skill_prerequisites` (the Skill Graph). The **roadmap is not a third structure** — it is the target role's required subgraph, minus what the student has demonstrated, layered by topological sort. Three views, one table.

Skills and roles



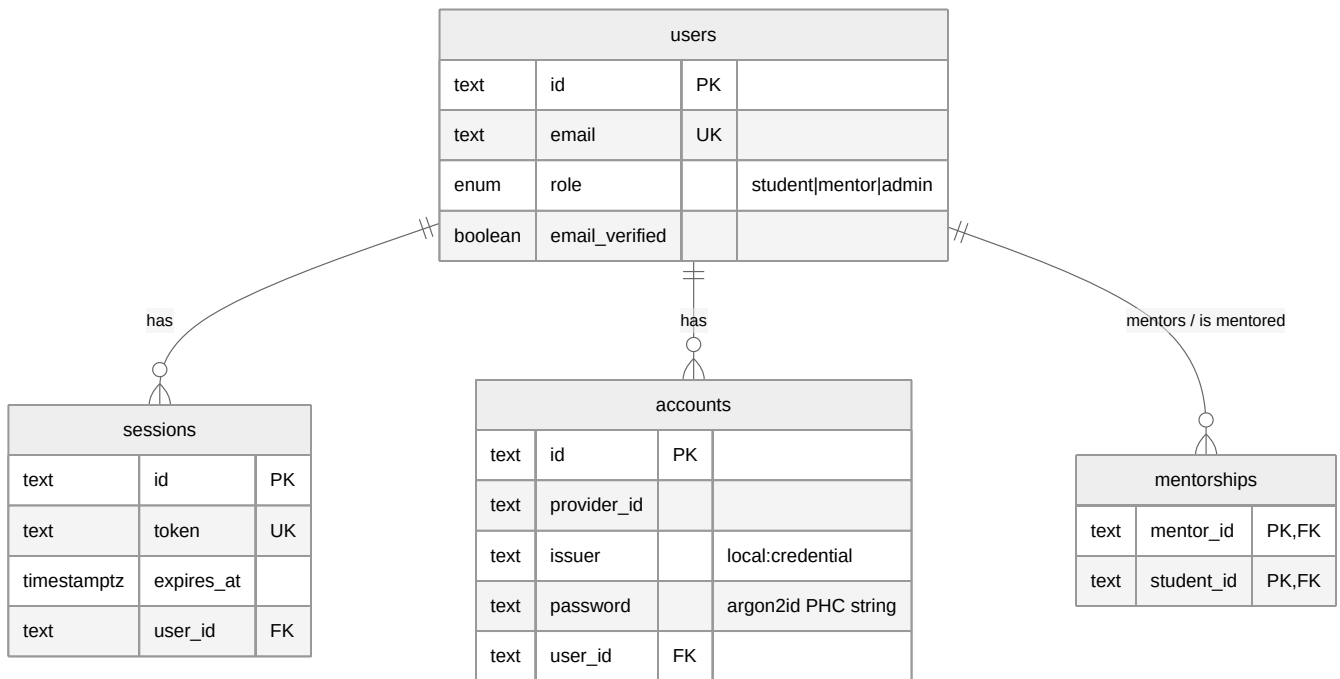
Two invariants cannot be expressed in SQL and are enforced on write, with `SkillGraph.validate()` in the Python service as the backstop:

- prerequisite edges connect leaves only (`altitude = 'SKILL'`)
- the prerequisite graph stays acyclic

The seed script checks both before it writes, with a depth-first cycle detector — an undefined topological sort would silently produce a nonsense roadmap.

`required_level` and `student_skills.level` share the 1–5 scale on purpose: the gap calculation is a subtraction, and `weight` is what stops every gap scoring the same.

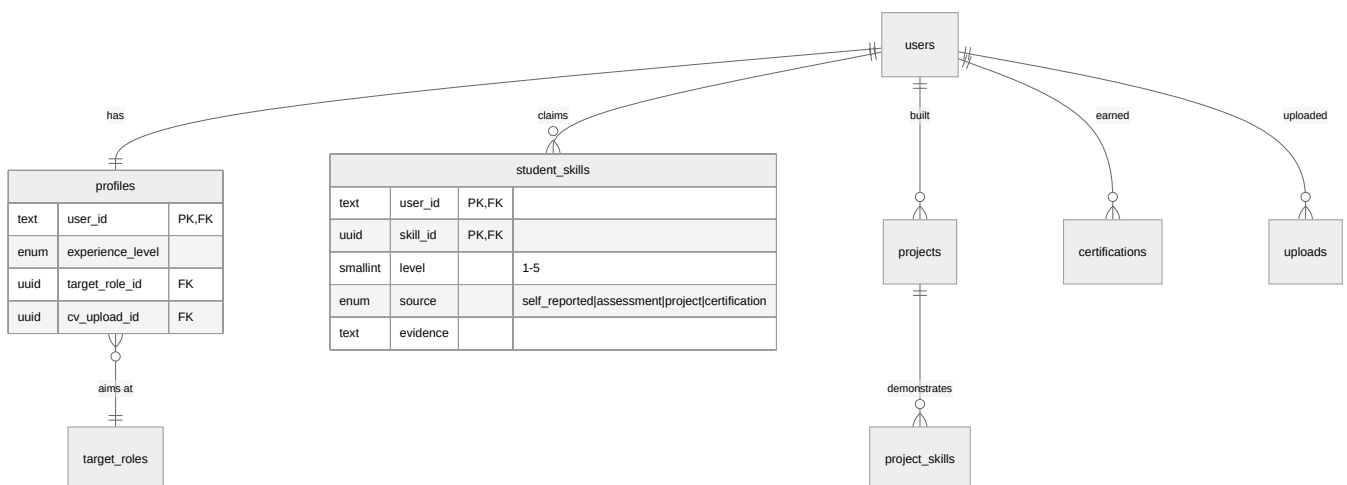
Identity



Better Auth's tables, adopted as source rather than regenerated. Four hand-applied edits are documented in `packages/db/src/schema/auth.ts` and must survive any regeneration: the `user_role` enum, `withTimezone` on every timestamp, the `rate_limits` table keeping its `id` column, and `accounts.issuer`.

The password hash lives on `accounts`, never on `users`. `mentorships` exists so a mentor's read access is a join, not a role-string comparison.

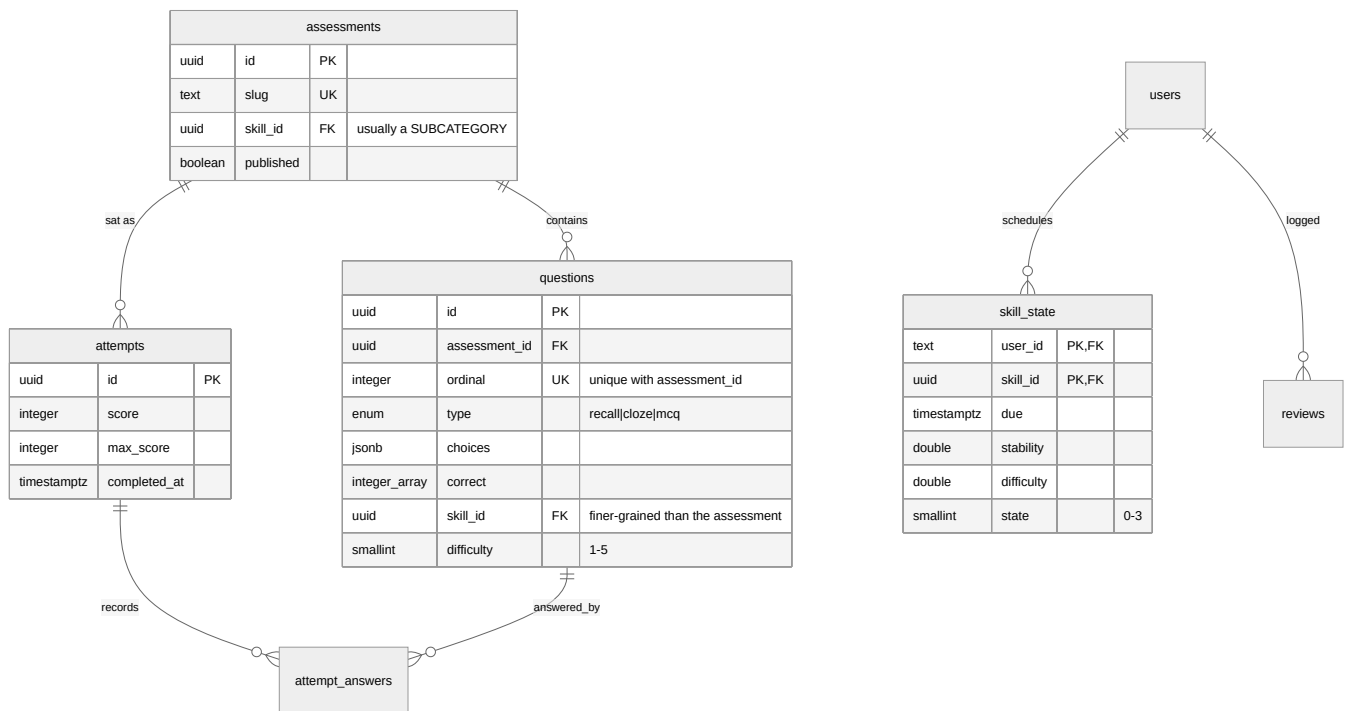
Student evidence



`source` is why the roadmap can weigh a claim differently from a result. An assessment overwrites `self_reported` and earlier `assessment` rows and leaves `project` and `certification` rows

untouched.

Assessment and progress

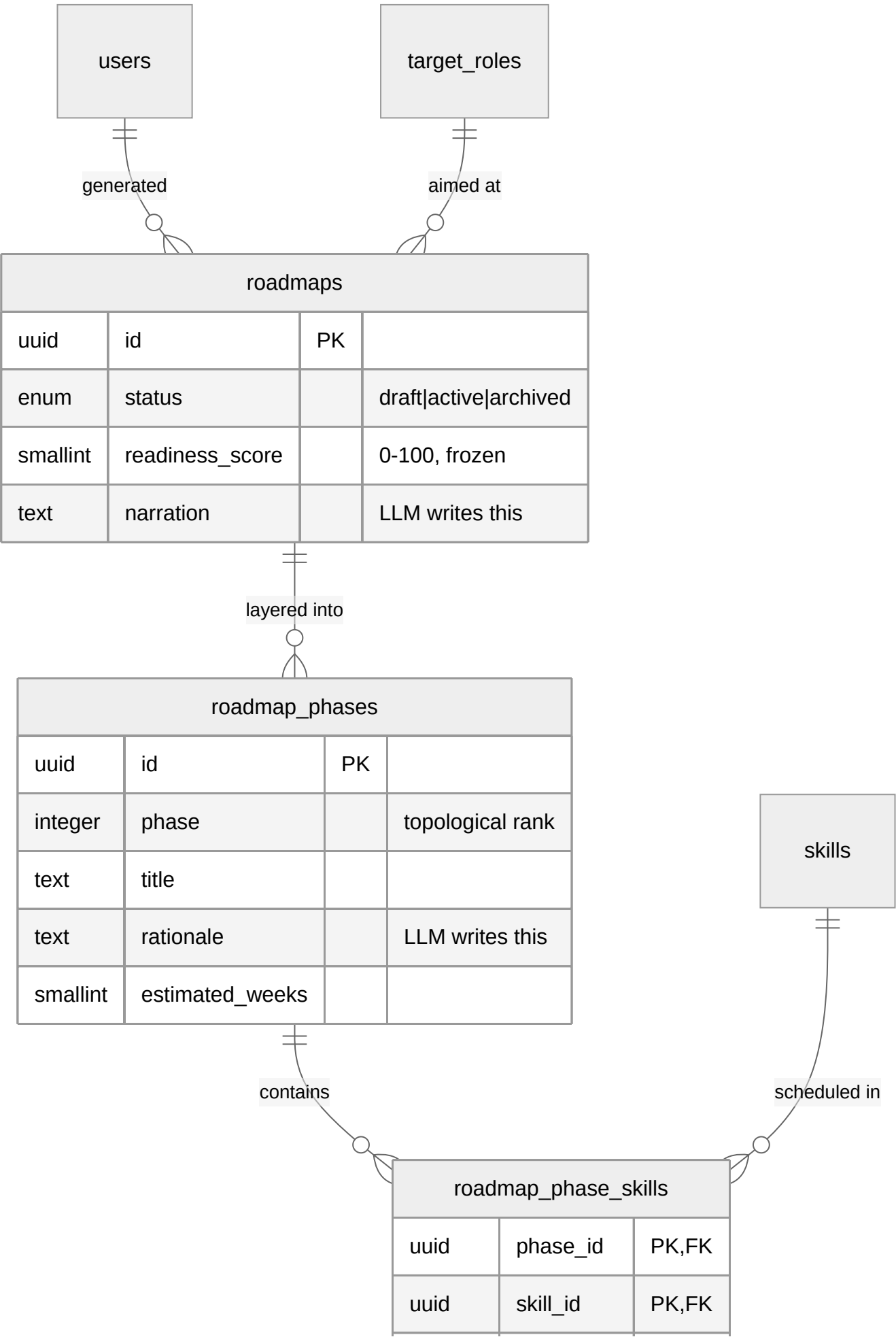


`questions.skill_id` being finer-grained than `assessments.skill_id` is what lets one sitting produce a per-skill breakdown instead of a single score.

`skill_state` is FSRS keyed on a **skill**, not a flashcard. Every column is needed to reconstruct a `ts-fsrs` Card; `state` in particular selects which scheduling branch runs. Proficiency decays and has to be re-earned.

The `(assessment_id, ordinal)` unique index is the natural key the seed upserts on, so re-seeding updates a question in place instead of orphaning the attempt answers that reference it.

Roadmap



integer	ordinal	
smallint	gap_score	

Stored output, not a source of truth. Python computes `phase` and ordering; the LLM only fills `narration` and `rationale`. `readiness_score` is frozen at generation rather than computed on read, so regenerating later shows movement instead of overwriting the evidence of it.

Making changes

Follow `.claude/skills/db-change/SKILL.md`. Typechecking does not prove a check constraint is valid SQL — every migration is verified against a real Postgres, and the constraint is proven to bite by inserting a row that should fail.

SkillForge · generated from `docs/schema.md`. The markdown source is authoritative.